MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Quiz 1 · Week 1: limits (Sections 1.5–1.7)**

8

15 minutes. 10 points. No notes, no calculators, no phones. Write your answers in the spaces provided and show all your work. Write "DNE" if a limit does not exist.

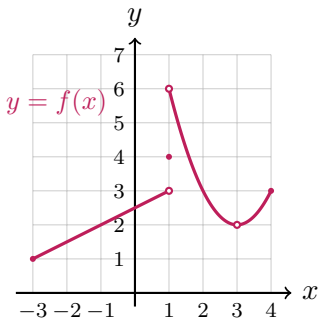
Name: _____

ID: _____

1	2	3	Total

Problem 1 *Reading a graph.* **[4 points]**

The graph of a function f is shown. Find each of the following.



(a) $\lim_{x\to 1^-} f(x) =$ _____

(b) $\lim_{x\to 1^+} f(x) =$ _____

(c) $\lim_{x\to 1} f(x) =$ _____

(d) $\lim_{x\to 3} f(x) =$ _____

Problem 2 *From the lectures.* **[2 points]**

(Question 3 of Lecture 2, word for word.) Circle one letter, and show the calculation that gives it.

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = ?$$

A. Substituting gives 0/0, which is undefined, so there is no limit.
B. 2
C. The limit does not exist because the function is undefined at 1.
D. 1

Calculation:

Problem 3 *An ε–δ proof.* **[4 points]**

Prove from the definition that $\lim_{x\to 2}(4x-3) = 5$. Scratch work on the left, the proof (written forward) on the right.

Scratch work — find δ

Proof